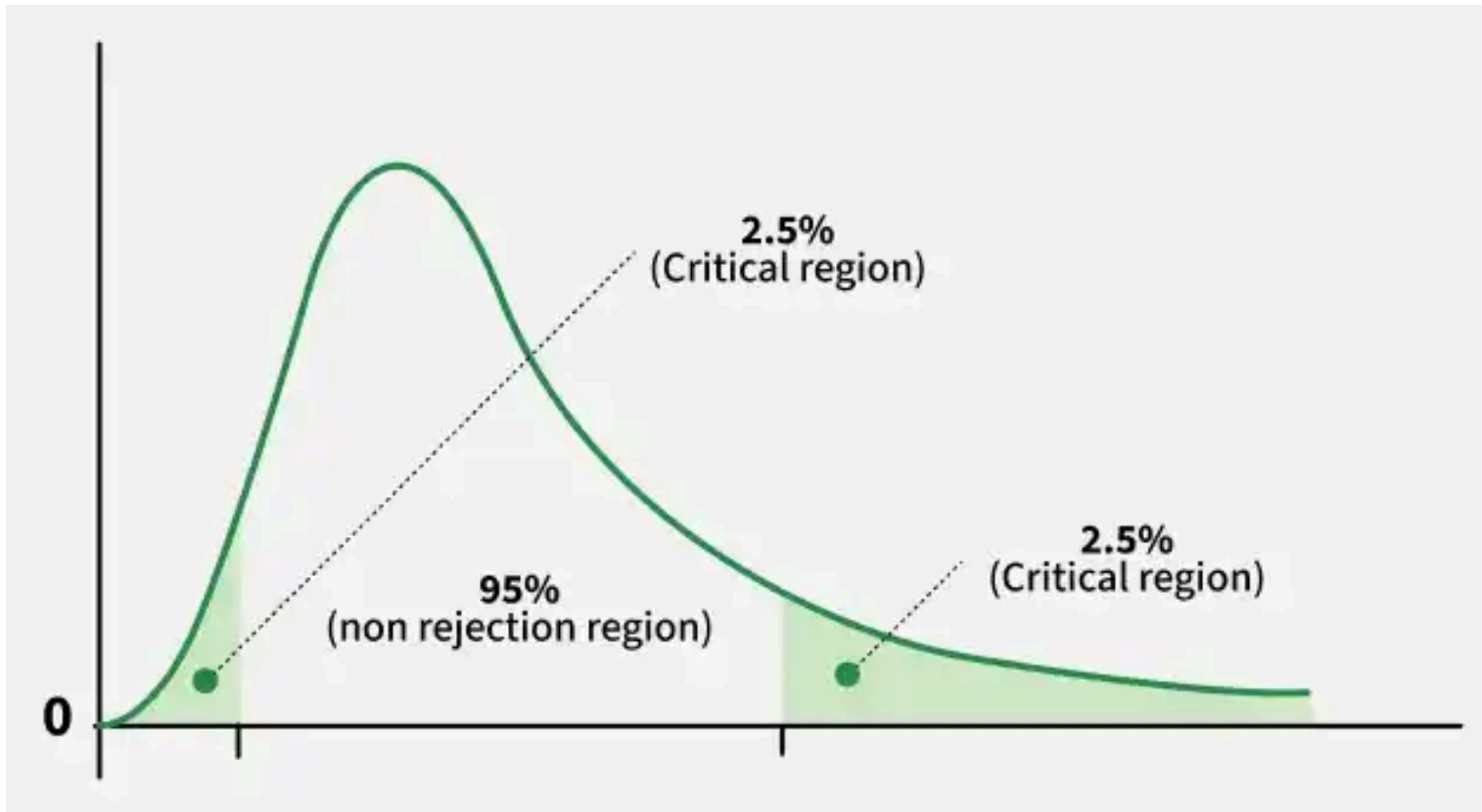


Chi-Square Test (χ^2 Test)



Suppose Amazon wants to know whether customer gender influences the purchase of electronic products.

Gender	Purchased
Male	Yes
Female	No
Male	Yes
Female	Yes

Can we use a Z-Test?

Why?

Because

- Gender → Categorical
- Purchase → Categorical

We are not comparing averages.

We are comparing frequencies (counts).

Hence, we use the **Chi-Square Test**.

What is the Chi-Square Test?

The **Chi-Square (χ^2) Test** is a statistical test used to determine whether there is a **significant difference between observed and expected frequencies**.

In simple words,

It checks whether what we observed is different from what we expected **by chance**.

Where is Chi-Square Used?

- Customer Purchase Analysis
- Fraud Detection
- Healthcare
- Survey Analysis
- Recommendation Systems
- Feature Selection
- Manufacturing
- HR Analytics

Real-World Examples

Healthcare

Does smoking affect lung cancer?

Variables

- Smoking
- Cancer

Both are categorical.

Banking

- Does loan approval depend on gender?

E-commerce

- Does payment method affect purchase completion?

Manufacturing

- Is machine failure related to work shift?

Education

- Does teaching method affect pass/fail rate?

Types of Chi-Square Tests

1. Goodness of Fit Test

Checks whether observed data follows an expected distribution.

Example

A company expects equal sales across four product categories.

Expected Sales

Observed

Category	Expected
A	25
B	25
C	25
D	25

Category	Observed
A	30
B	18
C	27
D	25

Do these observations match the expected distribution?

2. Test of Independence

Checks whether **two categorical variables are associated**.

Example

Gender	Purchased
Male	Yes
Female	No

Question

- Is purchasing behavior dependent on gender?

This is the most commonly used Chi-Square Test in Data Science.

Null and Alternative Hypotheses

For Independence Test

H_0 • The variables are independent. • (No relationship)	H_1 • The variables are dependent. • (There is a relationship.)
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Important Terminology

Before calculating the Chi-Square statistic, you should understand two important concepts.

1. Observed Frequency (O)

Observed frequency refers to the actual data collected from the sample.

Example:

Gender	Purchased
Male	40
Female	30

These values are the observed frequencies.

2. Expected Frequency (E)

Expected frequency represents the values we would expect if the null hypothesis were true.

If there is no relationship between the variables, what counts should we expect?

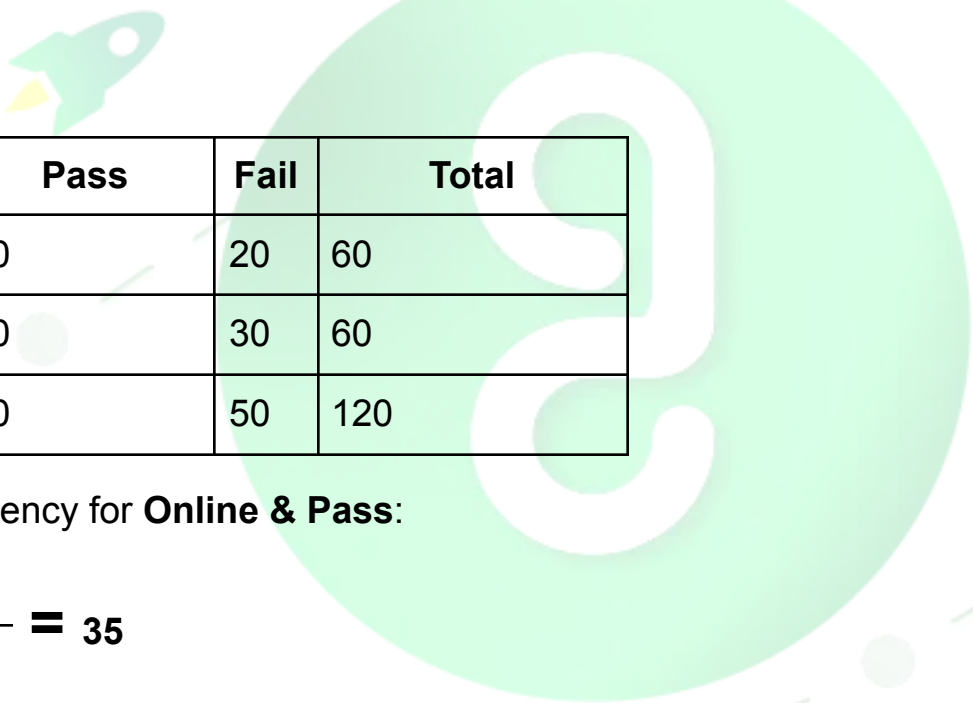
These expected frequencies are calculated mathematically using the row totals and column totals.

Formula for Expected Frequency

$$\text{Expected Frequency} = \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total}}$$

Example

Suppose



Training	Pass	Fail	Total
Online	40	20	60
Offline	30	30	60
Total	70	50	120

Expected frequency for **Online & Pass**:

$$\frac{60 \times 70}{120} = 35$$

Repeat the same calculation for each cell.

Chi-Square Formula

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

χ^2 = chi squared

O_i = observed value

E_i = expected value

The formula measures how far the observed frequencies are from the expected frequencies.

The Chi-Square Test is a powerful statistical technique used to analyze **categorical data**. Unlike the Z-Test and T-Test, which compare numerical means, the Chi-Square Test compares **observed frequencies** with **expected frequencies** to determine whether a statistically significant relationship exists between variables. It plays a crucial role in Data Science, Machine Learning, healthcare, finance, marketing, and social science research, making it one of the most important hypothesis tests for categorical data analysis.

Chi-Square Test of Independence

Problem Statement

An e-commerce company wants to determine whether **customer membership status** is associated with the **payment method** used.

They collected data from **200 customers**.

The observed data is shown below.

Membership Status	Cash	Card	Total
Member	50	70	120
Non-Member	30	50	80
Total	80	120	200

Step 1: Define the Problem

The company wants to answer the following question:

Does membership status influence the customer's payment method? or

Are the two variables independent?

The two categorical variables are:

- Membership Status (Member, Non-Member)
- Payment Method (Cash, Card)

Since both variables are categorical, we use the **Chi-Square Test of Independence**.

Step 2: State the Hypotheses

Null Hypothesis (H_0)

- Membership status and payment method are **independent**.
- There is **no relationship** between them.

Alternative Hypothesis (H_1)

- Membership status and payment method are **dependent**.
- There **is a relationship** between them.

Step 3: Observed Frequencies (O)

These are the actual values collected from the survey.

Membership	Cash	Card
Member	50	70
Non-Member	30	50

These are called the **Observed Frequencies** because they are directly observed from the data.

Step 4: Calculate the Expected Frequencies (E)

Now ask the learners:

If membership status has NO effect on payment method, what values would we expect?

This is exactly what the **Expected Frequency** represents.

The formula is:

$$E = \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total}}$$

Cell 1: Member – Cash

Cell 2: Member – Card

Cell 3: Non-Member – Cash

Cell 4: Non-Member – Card

Step 5: Expected Frequency Table

Membership	Cash	Card
Member		
Non-Member		

These values represent what we would expect if there were no relationship between membership status and payment method.

Step 6: Compare Observed vs Expected

Membership	O	E
Member–Cash	50	48
Member–Card	70	72
Non-Member–Cash	30	32
Non-Member–Card	50	48

Step 7: Chi-Square Formula

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

χ^2 = chi squared

O_i = observed value

E_i = expected value

Explain each part:

- **O** = Observed Frequency
- **E** = Expected Frequency
- **O – E** = Difference
- **(O – E)²** = Square of the difference
- Divide by **E**
- Add all values

Step 8: Manual Calculation

Cell	O	E	O-E	(O-E) ²	(O-E) ² /E
Member–Cash	50	48			
Member–Card	70	72			
Non-Member–Cash	30	32			
Non-Member–Card	50	48			

Cell	O	E	O-E	(O-E) ²	(O-E) ² /E
Member-Cash	50	48	2	4	4/48 = 0.083
Member-Card	70	72	-2	4	4/72 = 0.056
Non-Member-Cash	30	32	-2	4	4/32 = 0.125
Non-Member-Card	50	48	2	4	4/48 = 0.083

Step 9: Add All Values

$$\chi^2 = 0.083 + 0.056 + 0.125 + 0.083$$

$$\chi^2 = \mathbf{0.347}$$

Step 10: Degrees of Freedom

Formula

$$df = (Rows - 1) \times (Columns - 1)$$

Rows = 2

Columns = 2

$$df = (2-1)(2-1) = 1$$

Step 11: Find the p-value

Now we have

- $\chi^2 = 0.347$
- $df = 1$

Using a Chi-Square table, the p-value is:

$$p \approx 0.556$$

Step 12: Decision

Given

- $\alpha = 0.05$

Compare:

p-value	α
0.556	0.05

Since

$0.556 > 0.050$

Fail to Reject H_0

Step 13: Business Interpretation

There is **no statistically significant evidence** to conclude that membership status affects the payment method.

The small differences between the observed and expected counts are likely due to random chance.